Hypertension Monitoring Platform — Project Spec

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Full explanation of the organisers' brief: The brief explained

Overview

We turn the months of silence between two consultations into a continuous record of home blood pressure readings, and show the doctor only the patients who need attention today.

The problem. About 7.7 million people in Algeria have hypertension, around 6.4 million of them uncontrolled (figures from the hackathon brief). Doctors see one reading per visit, so a rising trend, a stopped treatment or a crisis between visits goes unnoticed.

The solution. A mobile web page for patients to record readings, pills and symptoms; an SMS channel for patients without a smartphone or internet; and a doctor dashboard that sorts patients by risk, explains every alert and tracks the brief's indicators.

Principle. The system assists; the doctor decides. It never diagnoses or changes a treatment.

Pitch line: "The guidelines decide, the doctor stays in charge, and no patient is left out because of their phone."

What we lead the pitch with:

1. **Missed-pill pattern detection:** "readings 15 mmHg higher on days with a missed pill".
2. **SMS-first access** for patients who will never install an app.
3. **Clinical credibility:** confirmation re-measure, white-coat comparison, silence detection.
4. **One slide of numbers:** the four indicators the brief names.

Name: TensioCare.

Users and roles

Two roles share one backend for the hackathon; each sees only what it needs.

Role	Uses	Main job	Sees
Patient	Mobile web page, or SMS	Record readings, pills and symptoms; follow advice	Own readings, status, advice, doctor messages
Doctor	Web dashboard	Review prioritised alerts, follow patients, prepare consultations	Own patients only, charts, alerts, indicators

A patient is linked to one doctor through an invite code the doctor gives at the consultation. A pharmacist role (screening, entering readings for patients without a device) is on the roadmap, not in the hackathon build.

Patient mobile web page

A single mobile web page, not a native app: faster to build, nothing to install, and a judge can open it on their own phone. Every reading takes under 30 seconds, in French or Arabic (right-to-left layout).

Screens

1. **Sign in:** phone number and code (fixed demo code `1234`; real OTP is roadmap), consent to data use, doctor invite code, name, age, sex, treatment names, emergency contact, language.
2. **Home:** today's status as icon + colour + word (Bon, Attention, Urgent), never colour alone; big "Add reading" button; last 7 days mini chart.
3. **Add reading (one screen):**
   - Short guidance first: sit 5 minutes, arm at heart level, no talking
   - Big +/− steppers for systolic, diastolic and pulse (no keypad typing)
   - Two readings 1 minute apart; the page stores their average
   - On the same screen: pill taken or missed, and symptoms (headache, dizziness, nosebleed, chest pain, vision problems, confusion, weakness on one side, difficulty speaking)
4. **Result screen:** status and the approved advice message for this situation.
   - Any danger symptom → straight to "Call 14 now", no re-measure
   - High reading, no danger symptom → "Rest 5 minutes and measure again"
5. **History:** chart with the target zone shaded.

6. **Contact my doctor:** call or message the doctor.

Key behaviours

- Symptoms are always collected with the reading, never after, so a danger sign is never delayed by a re-measure.
- If the page has no internet, a "Send by SMS" button opens the SMS app pre-filled with the reading (`TA 145 92 78 MP`), so the reading still arrives.
- Large text, icons with words, high contrast, works on small and old Android phones.
- French ships first; Arabic with right-to-left layout second, if time allows.
- Native app with automatic SMS and photo reading (OCR) is roadmap.

Doctor dashboard

The doctor opens the dashboard and sees, in one glance, which patients need attention today and why.

Screens

1. **Triage list (home):** patients sorted by risk, each with icon + colour + word, last reading, 7-day home average, trend arrow, adherence %, and the reason for the flag. Refreshes every 1.5 seconds, so a new red reading appears almost instantly.
2. **Alert inbox:** open alerts with their explanation ("2 of the last 3 readings above 160/100"). Actions: acknowledge, message the patient, close.
3. **Patient page:**
   - Chart of systolic and diastolic over time, target zone shaded
   - Pills taken or missed and symptoms marked on the timeline
   - Missed-pill pattern ("+15 mmHg on days with a missed pill")
   - **White-coat comparison:** "home average 132/84 vs last office reading 148/92: possible white-coat effect" (or the reverse: possible masked hypertension)
   - **Silence flag:** "no readings for 9 days, expected 2 per week", shown as a grey Silent tag so it never looks like a worsening patient
   - One-paragraph summary written by the LLM from the computed numbers
   - Doctor notes, last office reading, personal target thresholds
4. **Pre-consultation report:** the patient page in a print layout (browser print), no PDF generation.

5. **Indicators:** see the Indicators section.

Workload rules

- Only red alerts notify instantly; everything else waits in the triage list and a daily digest.
- Every alert says why it fired and on what evidence.
- The doctor can adjust thresholds for a specific patient.

Alert system

Danger symptoms are checked first and skip everything else; only high readings without symptoms go through confirmation. Every level states its evidence.

Step 1, danger symptoms first. Reading and symptoms are entered together. If any danger symptom is ticked (chest pain, vision problems, confusion, weakness on one side, difficulty speaking), the patient sees "Call 14 now" immediately, whatever the reading, with no re-measure.

Step 2, confirm (no danger symptom only). Each session is two readings 1 minute apart, averaged; this is the standard home measurement protocol. If the result is high, the page asks for one more session after 5 minutes of rest. This removes most false alarms.

Step 3, classify. Thresholds use the patient's own target set by the doctor, or the default (135/85 at home) when none is set. All values to be validated with a doctor.

Level	Evidence	Trigger	Patient sees	Doctor gets
Red, emergency	Any reading + a danger symptom	Danger symptom, whatever the reading	"Call 14 now" with a one-tap call button	Immediate alert; emergency contact notified
Red, crisis	Confirmed reading	180/120 or higher, no danger symptom	Rest, re-measure in 30 minutes, contact your doctor today	Immediate alert

Level	Evidence	Trigger	Patient sees	Doctor gets
Orange	2 of the last 3 readings	160/100 or higher, or below 90/60 with dizziness	Advice; "your doctor has been informed"	Top of the triage list
Yellow	7-day average or trend	Average above target (default 135/85), rising 14-day trend, or missed-pill pattern	Encouragement and advice	Daily digest
Grey, silent	Time since last reading	No reading for 2× the patient's expected interval (for example 7 days at 2 readings per week, 3 days for high-risk patients)	Reminder to measure	Grey "Silent" tag, separate from worsening patients
Green	7-day average	Within target	Bon, keep it up	Nothing; chart only

A single high reading. One unconfirmed reading above the target never raises an alert on its own. In the page it triggers the re-measure; by SMS the reply asks the patient to rest 5 minutes and send again, echoing the exact numbers received.

Step 4, escalation. A red alert not acknowledged within 30 minutes is sent to the doctor again, then to a backup or on-call contact. The patient is never told to call 14 because the doctor did not answer; 14 is shown only for danger symptoms, always.

Emergency contact. On a red emergency, the emergency contact gets an SMS without clinical details: "A family member reported a danger sign in TensioCare and was told to call 14."

Demo mode. A setting shortens the 1-minute and 5-minute waits to a few seconds, with a visible "Demo mode: timers shortened" label on screen, so the full flow fits in the pitch honestly.

Intelligence

All detection is explainable rules and simple statistics; an LLM is used for exactly two narrow language jobs and never decides anything.

Detection

Detection	Method	Example output
Crisis and danger	Thresholds on a confirmed reading, plus danger symptoms	"185/122 confirmed: crisis level"
Repeated high or low	2 of the last 3 readings past a threshold	"2 of last 3 above 160/100"
Trend	7-day moving average and 14-day linear slope, projected forward	"Systolic rising +2 mmHg per week; likely above target in about 10 days"
Missed-pill pattern	Average on days with a missed pill vs days with the pill taken	"Readings 15 mmHg higher on days with a missed pill"
Silence	No reading for 2× the patient's expected interval (scheduled job)	"Silent: no readings for 9 days, expected 2 per week"
White-coat or masked	7-day home average vs last office reading	"Home 132/84 vs office 148/92: possible white-coat effect"

The trend projection is the prediction feature: honest, explainable and accurate enough for a short horizon.

The two LLM jobs

1. **Rephrase an approved advice message** in simple French or Arabic. Darija is shown only as one pre-generated, labelled example ("Darija is spoken; the AI rephrases approved advice informally"). The message itself comes from an advice library reviewed against hypertension guidelines; the LLM only translates and simplifies it.
2. **Write the doctor's one-paragraph summary** from numbers the system already computed.

Hard limits: the LLM never chooses advice, never changes a dose, never diagnoses. The demo shows job 1 from a pre-generated response with a static fallback; the API is never called live on stage.

Reminders

Fixed reminders for the hackathon: morning and evening by SMS on measurement days. Adaptive frequency (more when something changes, less when stable) is roadmap.

Demo data

Synthetic Algerian patients, each with a story that triggers one feature:

- Fatima, 58, on amlodipine, stopped taking it 10 days ago: missed-pill pattern, orange
- Mohamed, 71, slowly rising over 3 weeks: yellow trend
- Karim, 49, office 148/92 but home average 131/83: white-coat comparison
- Zohra, 64, measures twice a week, no readings for 9 days: silence flag
- Ahmed, 62: the live red alert in the demo
- About 25 stable green patients for realism

Low-connectivity access

Any patient can send a reading without a smartphone or internet, which covers elderly and rural patients; this is our main differentiator for Algeria.

Channel	Needs smartphone	Needs internet	How
Mobile web page	Yes	Yes, or SMS button	Normal flow; without internet, a button opens the SMS app pre-filled
SMS	No	No	Patient sends `TA 145 92 78` (systolic, diastolic, pulse); add `MP` for a missed pill, or a symptom code such as `CP` for chest pain

SMS codes. `TA` = reading, `MP` = missed pill, `CP` = chest pain. Any danger symptom code triggers the "Call 14 now" reply immediately.

SMS replies. Every SMS gets an automatic reply that echoes the exact numbers received, never the patient's name. Example for `TA 165 100 78 MP`: "Tension elevee (165/100). Reposez-vous 5 minutes et renvoyez une mesure."

Hackathon setup. An Android phone running an SMS gateway app forwards incoming SMS to a webhook and sends the replies. Tested first, because Android may stop background SMS apps; a "paste incoming SMS" simulator is the backup. In production this becomes an operator short number.

Roadmap. Pharmacy entry and screening campaigns for early detection.

Indicators

The brief names four indicators; the dashboard shows exactly those four first, then two that prove the health outcome.

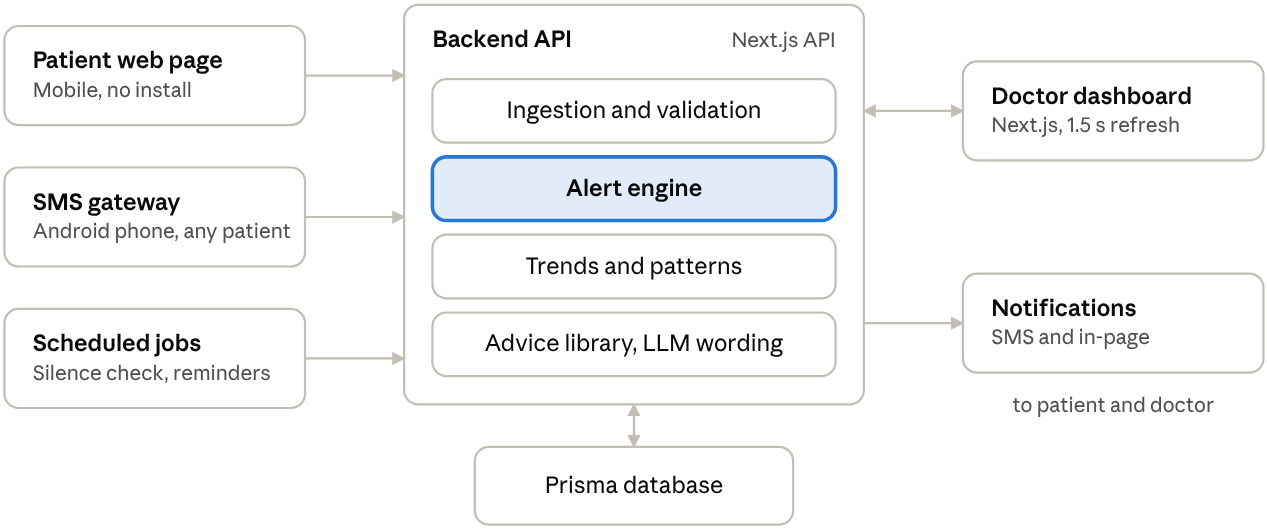
Indicator	Named in the brief	Definition
Measurement frequency	Yes	Readings per patient per week
Evolution of values	Yes	Group average systolic and diastolic over time
Number of alerts	Yes	Alerts per level per week
Regularity of follow-up	Yes	% of patients who measured on their expected days; patients flagged silent
Control rate	No	% of patients with a 7-day home average below 135/85
Response time	No	Median time from red alert to doctor acknowledgement

Each indicator is a big number plus a small trend chart, filterable by period. This page is one slide of the pitch.

Architecture and stack

One backend receives readings from the web page and SMS, runs the alert and analytics logic, and pushes results to the doctor dashboard and back to the patient.

Every channel feeds one backend that alerts the doctor



system architecture · 2 channels (web, SMS) + scheduled jobs, 1 backend, 2 outputs

Whatever the channel, a reading is normalised into the same format before the alert engine sees it, so adding a channel later means adding one adapter.

Stack

Part	Technology	Why
Patient page	Next.js mobile web page, Tailwind, French then Arabic (RTL)	Nothing to install; no native build to debug
Doctor dashboard	Next.js, Tailwind, TanStack Query (refetch every 1.5 s), Recharts	Polling looks live on stage, no WebSocket setup
Backend	Next.js API routes or Express, Zod validation, simple JWT with two roles	One codebase if using API routes
Database	Prisma with SQLite if the backend runs on one machine; hosted Postgres (Neon or Supabase) if deployed serverless	SQLite files do not persist on serverless hosting
SMS	Android phone with an SMS gateway app, forwarding to a webhook	Hours of work, real SMS in the demo

Part	Technology	Why
Scheduled jobs	Cron route every hour	Silence detection and reminders
LLM	Hosted LLM API, two narrow jobs only	Darija rephrasing, doctor summary
Auth	Fixed demo code `1234`	Real OTP is roadmap

Data model

Table	Key fields
User	id, role (patient, doctor), phone, name, language
Patient	userId, doctorId, age, sex, treatmentNames, targetSys, targetDia, emergencyContactPhone, expectedReadingsPerWeek, consentAt, lastOfficeSys, lastOfficeDia
Doctor	userId, backupContactPhone, digestTime
Reading	id, patientId, sys, dia, pulse, takenAt, channel (web, sms), pillTaken
Symptom	readingId, type (headache, dizziness, chest_pain, vision, confusion, weakness, speech, nosebleed)
Alert	id, patientId, level, evidence, reason, status (open, acknowledged, closed), createdAt, ackAt
Message	id, fromUserId, toUserId, body, createdAt
AdviceItem	id, situation, text, reviewedBy

Main API endpoints

- `POST /readings` from the page; `POST /sms/inbound` from the gateway
- `GET /doctor/patients?sort=risk`, `GET /patients/:id`, `GET /patients/:id/readings`
- `PATCH /alerts/:id` to acknowledge or close; `POST /messages`
- `GET /indicators?from=&to=`; `POST /jobs/silence` run by cron

Hackathon MVP

One loop must work perfectly on stage: a reading enters the system, a dashboard row turns red in under 2 seconds, the doctor clicks it, sees why, acknowledges, and messages the patient. Everything else is secondary.

First hours, in this order

1. Deploy the dashboard to a real URL, not localhost.
2. Test the SMS gateway phone; if it fails by hour 4, switch to the "paste incoming SMS" simulator and keep the phone for the demo only.
3. Build the core loop end to end before any extra feature.
4. Record a 90-second backup video of the full loop before the final pitch.

Must have

- [ ] Patient web page: sign in with consent, add reading with steppers, two-reading average, pill and symptoms on the same screen, result screen, history chart
- [ ] Danger symptoms skip confirmation and show "Call 14 now"
- [ ] Demo mode with shortened timers and a visible label
- [ ] SMS channel through the gateway phone, with automatic reply echoing the numbers
- [ ] Alert levels with evidence rules, patient target or default threshold
- [ ] Doctor dashboard: triage list (1.5 s refresh), alert inbox with explanations, patient page with chart, acknowledge and message
- [ ] Trend, missed-pill pattern, silence (grey tag), white-coat comparison
- [ ] Indicators page with the brief's four indicators
- [ ] Algerian synthetic patients seeded
- [ ] Icon + colour + word on every status; French interface

Only if time remains

- [ ] Arabic interface with right-to-left layout
- [ ] One pre-generated Darija rephrasing of an advice message, labelled as an example
- [ ] Emergency contact SMS on red emergency
- [ ] Doctor summary paragraph by the LLM (pre-generated for the demo)
- [ ] Print layout for the pre-consultation report

Roadmap (say it in the pitch, do not build)

Native app with automatic SMS and photo reading of the monitor, adaptive reminders, real OTP, pharmacy entry and screening, operator short number, Bluetooth monitors.

Team split (4 people)

Person	Owns
1	Patient web page
2	Doctor dashboard and indicators
3	Backend, database, SMS gateway, alert rules
4	Detection logic, seed data, advice library, pitch, backup video

With 3 people, merge roles 3 and 4. If the hackathon is under 36 hours, drop everything in "Only if time remains".

Demo scenario (about 3 minutes, demo mode on)

1. Doctor dashboard: 30 patients sorted by risk. Open Fatima: "readings 15 mmHg higher on days with a missed pill". Open Karim: home vs office, possible white-coat effect. Point at Zohra's grey Silent tag.
2. On a phone, Ahmed enters 182/118 with the steppers, no danger symptom; the page asks to re-measure (seconds in demo mode); the second session confirms.
3. On screen, Ahmed's row turns red in under 2 seconds. Click: "Confirmed 182/118, crisis level". Acknowledge, send "Come to the clinic today".
4. A basic phone with no internet sends `TA 165 100 78 MP` by SMS; the reading appears on the dashboard, and the phone gets a reply echoing 165/100 and asking for a second reading.
5. End on the indicators page: measurement frequency, evolution, alerts, regularity.

Questions to expect from the jury

- Who validated the thresholds? Guideline values, to be validated with a partner doctor; each patient can have a personal target.

- What if the patient has chest pain? Danger symptoms skip everything and show "Call 14 now" at once.
- What if the doctor does not answer a red alert? Second notice, then a backup contact; 14 is only for danger symptoms.
- Where is the AI? Only for language: it rephrases approved advice and summarises numbers; it never decides.
- How is patient data protected? Consent at sign-up; each doctor sees only their own patients; SMS replies never contain the patient's name and patients are matched by phone number; readings stored as pseudonymised records; designed to be hosted in Algeria in line with Law 18-07 on personal data (a design intent, not a legal claim).